Compressing physical properties of atomic species for improving predictive chemistry

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Introduction

One of the most important aspects of good machine learning models is the representation of data in a form which is suitable for the algorithm. In the chemical sciences this often means representing atomic species by discrete representations such as atomic number, or by using one-hot channels for each unique atomic species, neither of which are ideal solutions. In this work we derive a continuous vector representation by compressing properties of each element with an auto-encoder, which captures underlying physical law and provides many advantages to several neural network models shown here.

Python code for this work is available at https://github.com/jeherr

Deriving a vector representation from physics

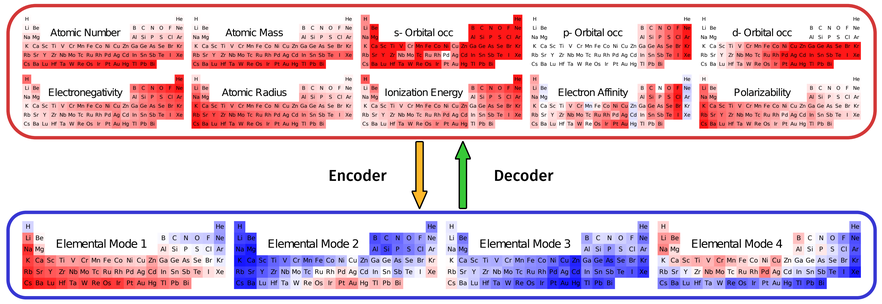

Figure 1: A set of ten physical properties for each element is compressed with an auto-encoder into a vector representation which captures the underlying physics. We refer to this representation as the elemental modes.

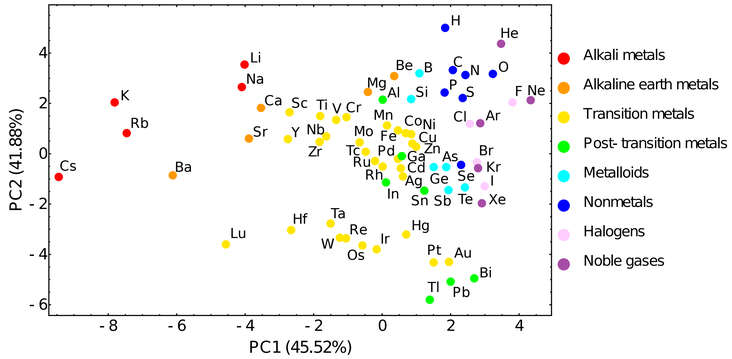

Figure 2: Principal component analysis on the elemental modes shows a striking similarity to the periodic table, showing our representation has retained essential information about the elements.

Predicting formation energies

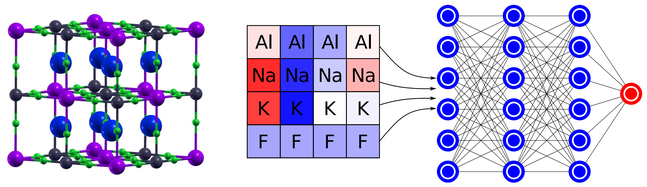

Figure 3: We used the elemental modes as the feature vector to a neural network to predict formation energies of elpasolites, crystalline materials of the formula ABC_2D_6.

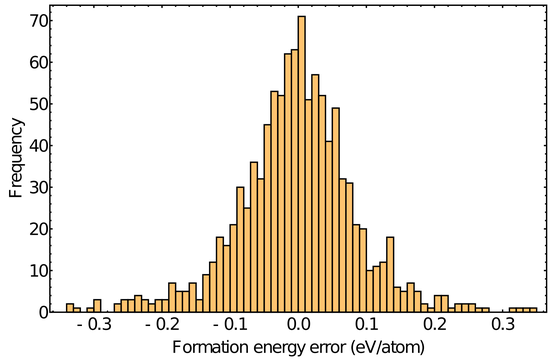

Figure 4: Histogram of predicted formation energy errors for an independent test set. MAE was 67 meV/atom. Previous works on this same task achieved MAEs of 100 meV/atom.

High-dimensional neural network potential

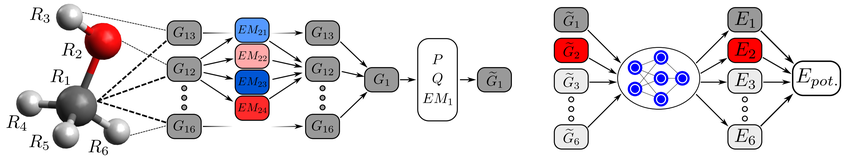

Figure 5: Schematic representation of HDNNP scheme in our work. The elemental modes are used to scale the symmetry functions across four channels to represent the identity of neighboring atoms. We introduce a tensor decomposition with two learnable matrices (P and Q) and the elemental modes of the central atom to encode its own identity into the feature vector. This enables us to keep the feature vector constant size and to use a single neural network branch for each atomic species. Our data set consisted of 1.35 million geometries from 65,000 unique molecules from the Chemspider database. RMSE of an independent test set for the energy and forces is 0.0976 kcal/mol per atom and 3.71 kcal/mol/Å respectively.

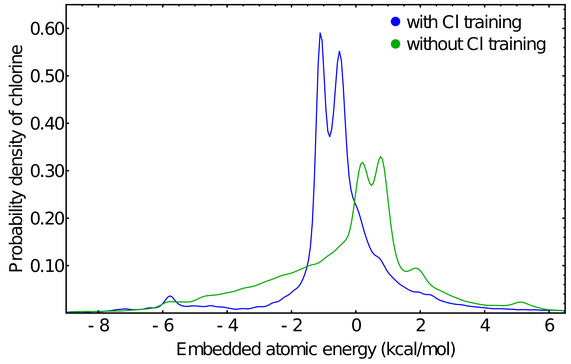

Figure 6: Training another HDNNP without any Cl containing molecules, the model makes reasonable predictions similar to those from the model trained with the full data set. By using a single neural network branch for each atomic species, our model shares information learned about distinct atomic species to improve predictions. The elemental modes allows us to reduce the need for training data and to improve prediction accuracy. Further, modeling new elements does not require retraining our model from scratch.

Alchemical transformations

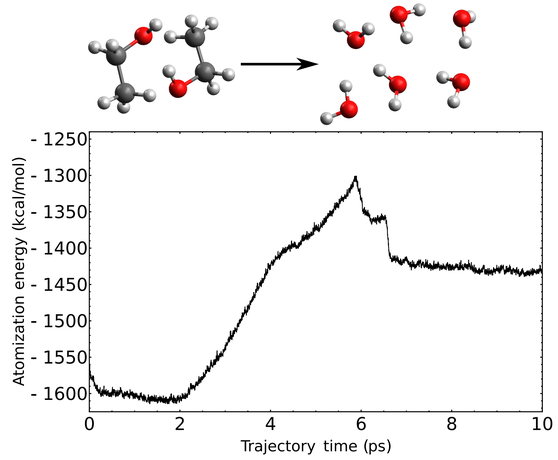

Figure 7: Our HDNNP scheme enables us to define and evaluate feature vectors for intermediate states of an alchemical transformation and to perform alchemical free energy calculations. Above we show an MD trajectory of an ethanol dimer transforming into a water hexamer.

Conclusion

Our vector representation has been shown to be superior to other representations of atomic species on similar modeling tasks for predicting formation energies. When used in conjunction with the symmetry functions and HDNNPs, it allowed us to keep computational cost down, reduce the need for training data, and open up the possibility for neural network potentials to perform alchemical free energy calculations.

References

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